

MAPLE (FENGYI) ZHANG

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EDUCATION

University of Illinois Urbana-Champaign

Master of Computer Science

Aug. 2022 - Dec. 2024

GPA: 3.56/4.0

Research Area: Data Systems, Machine Learning, Autonomous Vehicle, Cloud Networks, Internet of Things

Shandong University

Bachelor of Engineering in Computer Science and Technology

Sept. 2017 - Jun. 2021

Overall GPA: 3.82/4.0

Relevant Coursework: Data Structure & Algorithm, Computer Networks, Database Systems, Operating Systems

TECHNICAL SKILLS

Programming: Go, Python, TypeScript/JavaScript, C/C++/C#, Java, PHP, SQL, Solidity, Shell

Frameworks/SDKs: React, Next.js, Node.js, FastAPI, Gin, React Native, Express, Prisma, GORM, gRPC, REST, GraphQL, PyTorch, .Net, Tailwind CSS

Cloud/Platforms: Git, GitHub Actions, Docker, Kubernetes, AWS (ECS, Lambda, RDS), GCP, Azure, Vercel, PostgreSQL, MySQL, MongoDB, SQLite, Qdrant, Redis, Celery, Nginx, Pulsar, Firebase, Jenkins

Data/Quality: Pandas, NumPy, ECharts, Cesium, H3, Jest, Playwright, testing, observability

AI / Agent Engineering: Codex, Claude Code, Cursor, GitHub Copilot, OpenClaw, MCP, Agent Skills, multi-agent orchestration, RAG, vector databases, prompt engineering, LangChain, Anthropic API, OpenAI SDK

PROFESSIONAL EXPERIENCE

Space AIC

Mar. 2026 – Present

Software Engineer, Shanghai, China

- Built frontend capabilities for the world's first 10,000+ satellite digital-twin system with React 19, TypeScript, Vite, Cesium, Zustand, and ECharts; expanded supported experiments from 1,800 to 15,000+ satellites while keeping 3D orbit/attitude playback interactive.
- Integrated the React/Cesium control plane with a distributed RPC simulation backend spanning topology generation, weighted path computation, and sharded satellite simulators; normalized result frames for constellation design, attitude control, interference, thermal, power/energy, space-environment, and ns-3 experiments.
- Delivered hybrid-in-the-loop workflows with microVM terminals and a Terminal Gateway over HTTP/WebSocket/gRPC, including entity/path terminal routing, resource monitoring, draggable multi-window management, and session-aware state handling.
- Built high-performance Cell/Beam rendering for 1,584-satellite experiments: drew H3 grids, terrain, and 16 beam cones; displayed satellite-wide footprints or regional coverage in real time with automatic tracking and switching.
- Developed SAT Agent for satellite Q&A and simulation support, a digital-twin copilot for scenario analysis, and a space-ground edge-AI architecture for on-satellite large-model inference.
- Built SDT Education's Notebook—Python engine—Express API—Cesium workflow and per-student JupyterHub/Docker labs; authored ground-to-satellite, inter-satellite-link, end-to-end transmission, and shortest-path selection experiments to close the model—code—simulate—visualize—submit—grade loop.

Nethermind

Aug. 2023 – Nov. 2023

Internship Program @Ethereum Foundation, Remote

- Implemented and migrated new architecture design for Voyager, a StarkNet blockchain explorer, from AWS Lambda and Starknet gateway to AWS ECS and RPC endpoints, rewriting 2 APIs and drawing 2 arch diagrams
- Completely refactored search module to provide user appropriate search options depending on their input, resulting in a reduction of search latency by 1000 ms
- Optimized and improved retrieving efficiency from AWS RDS by adding indices and rewrites postgresql queries
- Devised an prompt engineering method for auto-detecting and decoding Cairo types to display transaction call-data in human-friendly format, refining accuracy wrt. traditional regex matching.

Microsoft

Jan. 2022 – Aug. 2022

Software Engineer, Suzhou, China

- Designed and implemented a seamless PC-Phone pairing flow via QR/Accessibility code with 8 RESTful APIs and .Net Framework, increasing completion rate during Windows OOBE from 28.3% to 35.4%.
- Reduced server and database pressure by employing microservice architecture, asynchronous programming, caching, websockets, long polling, and API throttling, achieving a 97% API success rate at 720 RPS.
- Built account-based cryptographic trust relationships between PCs and phones by leveraging JWT encryption on certificate, device, and account information to ensure secure data transportation and verification.
- Improved service reliability and availability by adding over 2,000 lines of unit/functional/end-to-end/load tests and metric telemetry for anomaly detection, analytics, and troubleshooting.

Didi Global

Jan. 2021 – Dec. 2021

Software Development Engineer, Beijing, China

- Refactored 6 APIs of Point of Interest (POI) service that serves 15 million users per day from PHP to Go, among which CPU idle time improved to 73%(36% BEF) and TP99 latency reduced by 51% given QPS of 2000
- Divided business process into pluggable modules with inner workflow engine and replaced 8 RESTful APIs with Didi RPC (DiRPC) protocol with Service Discovery (DiSF) components for better SOA governance
- Applied A/B testing to realize canary deployment, configuration synchronization and strategy experiment, and employed metrics components to trace logs generated during service context for trouble shooting
- Managed online metric alarms (error ratio, latency etc.) of over 22 APIs and 16 downstream services while on-call

Alibaba Cloud

Jul. 2020 – Sept. 2020

Software Engineer Intern, Hangzhou, China

- Chaired project of route traceback based on by-pass sniffing for SLA monitoring and network fault location
- Constructed spoofed source VIP packet sequences with increasing TTL fields to specific endpoint using Scapy
- Employed traffic mirroring service for packets capturing and asynchronous I/O with call-back filtering function
- Delivered on Advanced Anti-DDoS clusters while being elastically extendable via configuration files

PERSONAL PROJECTS

Time Machine - Emotional Companion App

Jan. 2026 – Mar. 2026

SecondMe digital-twin companion with temporal memory distillation/RAG and dual-agent SSE chat

- Built and deployed a full-stack emotional companion around a SecondMe digital-twin Agent with **Next.js 16**, App Router, Prisma/SQLite, OAuth2, and Fly.io persistent volumes, giving users a private way to revisit conversations anchored to a chosen date.
- Engineered temporal memory distillation/RAG by extracting dated anchors from text and images, then injecting only memories before the chosen time point, preventing present-day context from leaking into simulated past conversations.
- Implemented dual-agent real-time **SSE** streaming: SecondMe handled primary dialogue while an Anthropic Act healer agent produced structured emotional analysis and empathetic responses.
- Designed a Canvas/requestAnimationFrame TimePortal state machine from onboarding through relationship and memory selection to time travel and conversation, turning a complex flow into a coherent emotional experience.

Type Machine (Media Transcriber)

2026

AI-native multimodal capture and knowledge-processing PWA

- Built a mobile-first **Next.js App Router** PWA with Web Share Target ingestion, letting users send screenshots and videos directly from the system share sheet or upload images, audio, video, and URLs from desktop.
- Designed an asynchronous multimodal job pipeline that routes media through OCR, speech transcription, and multimodal-model processing, turning heterogeneous inputs into a consistent editable-note schema without blocking uploads.
- Implemented ordered screenshot reconstruction with overlap deduplication, UI-noise filtering, language detection, and source-text preservation, eliminating the manual OCR copy-and-stitch workflow for long articles.
- Added AI post-processing for summaries, tags, and bilingual output, plus persisted task history, search/edit/review workflows, and Flomo API delivery so processed knowledge remains traceable and reusable.

OnePoem

Apr. 2026

Cross-platform poetry composition and export tool

- Created a mobile-first single-page poetry editor with **React**, reusable templates, typography, color, layout ratios, collections, and widgets, turning plain text into shareable visual pages.
- Implemented PNG export with html2canvas and an SVG fallback so compositions remain exportable across browsers and constrained mobile runtimes.
- Packaged the web experience as a **SwiftUI/WKWebView** iOS app and bridged JavaScript messages to native Photos saving and system sharing.
- Designed reusable styles and collection workflows so users can preserve a personal visual language instead of rebuilding each poem from scratch.

Hang Seng Index Futures Quantitative Trading System

Oct. 2024

Multi-timeframe strategy, backtesting, and risk management

- Built a validated market-data pipeline integrating YFinance and external APIs for synchronized processing from one-minute bars to weekly data.
- Developed a Stochastic Index trading strategy combining peak–valley divergence and parallel-channel detection to confirm signals across multiple timeframes.
- Designed risk controls with dynamic position sizing, adaptive stop-loss, and trailing stops to constrain exposure under changing volatility.
- Implemented modular backtesting, real-time alerts, and automated execution with structured logging and fault handling, enabling the same strategy logic across research and live workflows.